PROJECT DOCUMENTATION

Global Olympic Sports Performance Analysis

A Tableau Data Visualization Project

Project Title	Global Olympic Sports Performance Analysis
Tool Used	Tableau
Primary Dataset	athlete_events.csv
Supporting Dataset	noc_regions.csv (NOC-to-region mapping, where used)
Institution	Medi-Caps University
Student Name	Vidhi Jain
Programme / Semester	B.Tech CSE / 3rd Semester

Prepared as a concise report describing the dataset, Tableau worksheets, dashboard, and interpretation of the visualisations. Add screenshots or links if your submission rubric requires them.

1. Project Overview

Global Olympic Sports Performance Analysis is a data visualization project built in Tableau. It explores historical Olympic athlete-event records and presents selected patterns using three worksheets combined into a dashboard. The current workbook includes a Season pie chart, a Gender Trend line chart, and a Top Countries horizontal bar chart.

1.1 Objectives

• Explore the distribution of recorded Olympic event entries across Summer and Winter seasons.

• Visualise changes in recorded event entries by sex across Olympic years.

• Compare the number of recorded event entries associated with the leading NOC codes.

• Combine the worksheets in a single dashboard for easier viewing and comparison.

1.2 Project Deliverables

Deliverable	Description	Status / note
Dataset connection	CSV data connected in Tableau.	Completed
Three worksheets	Season, Gender Trend, Top Countries.	Created
Dashboard	Combined view with all three worksheets.	Created
Story	A Tableau story point containing the dashboard.	Verify/create if required
Packaged workbook	Save as .twbx for easier sharing.	Verify saved extension
Documentation and demo	Report, screenshots, and demo recording/link.	Add if required

2. Dataset Description

The primary data source is **athlete_events.csv**, a historical Olympic athlete-event dataset. The connected Tableau data preview showed 271,116 rows and 15 fields in the version used for this project. The supporting file **noc_regions.csv** can be used to map National Olympic Committee (NOC) codes to region or country labels, depending on the available fields and the data relationship established.

Field	Meaning / use
ID	Athlete identifier; can be counted distinctly to estimate unique athletes in a chosen view.
Name	Athlete name.
Sex	Recorded sex category used for the gender-trend visualisation.
Team	Team name associated with the event entry.
NOC	National Olympic Committee code; it is not always identical to a country name.
Games / Year	Olympic edition and year; used for time-based analysis.
Season	Summer or Winter season; used in the pie chart.
Sport / Event	Sport and event information, useful for participation analysis.
Medal	Gold, Silver, Bronze, or missing when no medal was recorded.

Data note: Missing values in Medal are expected for entries where no medal was recorded. They should not be removed from participation analysis by default. For a medal-only chart, filter Medal to Gold, Silver, and Bronze.

3. Tableau Visualisations

3.1 Season — Pie Chart

Purpose: compare the number of recorded athlete-event entries in Summer and Winter Olympic seasons. The Season field defines the slices, while the record count defines their size. Suggested title: “Summer vs Winter Olympic Event Records.” Percentages may be displayed for readability.

Interpretation: A larger slice indicates more rows in the dataset for that season. It does not directly represent unique athletes or medal totals.

3.2 Gender Trend — Line Chart

Purpose: show how the number of recorded event entries varies over time for the categories in the Sex field. Year is used along the horizontal axis, with record count on the vertical axis and Sex used to distinguish the lines.

Interpretation: The chart presents historical event-entry trends. Changes may reflect changes in the Olympic programme, number of events, and dataset coverage, not only changes in individual athlete counts.

3.3 Top Countries — Horizontal Bar Chart

Purpose: compare event-record counts for NOC codes. NOC is placed on Rows and record count on Columns; a Top 10 filter limits the chart to ten entries.

Interpretation: Longer bars indicate more recorded entries for that NOC code. Use a validated NOC mapping if displaying country names. The current measure is a record count, not a medal count.

4. Dashboard Design

The dashboard titled **Global Olympic Sports Performance Analysis** combines the three worksheets on one screen. This layout gives viewers an overview of season distribution, historical gender trends, and leading NOC codes by recorded event entries. Keep filters and legends readable and ensure chart titles accurately describe each metric.

Dashboard quality checks

■ Confirm all three charts appear and are not clipped.

■ Use descriptive titles and readable axis labels.

■ Keep the Top 10 filter applied to the country/NOC chart.

■ Verify legends and filters behave as expected.

■ Save the workbook after final layout changes.

5. Story

A Tableau Story is a sequence of story points and is separate from a dashboard. If required, create a story named **Olympic Performance Story**, add the completed dashboard as a story point, and use this caption: “This dashboard presents an overview of Olympic season records, historical event-entry trends by sex, and the top NOC codes by recorded event entries.” Preview the story and save the workbook again.

Status reminder: Mark the Story deliverable as complete only after you have created and tested it in Tableau.

6. Testing and Validation

Check	What to verify
Data source	Workbook opens and the CSV data source is available.
Season chart	Both season categories are represented and labels are legible.
Gender Trend	Year is on the time axis and the Sex legend is visible.
Top Countries	Exactly ten NOC codes appear when the Top 10 filter is active.
Dashboard	All charts and the title are visible at the chosen dashboard size.
Story	If required, the story point opens and displays the intended dashboard.
Saved workbook	Reopen the saved .twbx/.twb file to confirm there are no broken data links.

7. Limitations

• The current charts use counts of rows in athlete_events.csv. These are athlete-event records, not automatically unique-athlete counts or medal totals.

• The Top Countries chart uses NOC codes; country labels should be mapped with noc_regions.csv if readable country names are required.

• The standard athlete_events.csv dataset does not include country-level sports investment figures. Investment-versus-medal-efficiency claims cannot be calculated from this file alone.

• Cite the original dataset source and confirm its licence before final submission.

8. Conclusion

This project demonstrates the use of Tableau to explore historical Olympic athlete-event records through three visualisations and a combined dashboard. The Season chart compares Summer and Winter records, the Gender Trend chart presents changes over time, and the Top Countries chart compares leading NOC codes. Accurate interpretation depends on understanding that row counts represent recorded entries. The project can be extended by using a verified NOC mapping, adding medal-specific calculations, and creating a Tableau Story if required.

9. Project Links and References

Item	Add your verified link / details
Dataset source	Paste the original Kaggle or approved dataset URL here.
GitHub repository	Paste the repository URL after uploading the project files.
Tableau Public workbook	Paste the published URL if publishing is available and required.
Demo video	Paste the accessible video link after recording and uploading.

Before submission: paste verified links and insert screenshots if your teacher requires them. Do not state that optional deliverables are complete until they have actually been created and tested.